i) Google Question ⟹ $A \% P$.

ii) Alternating Subarrays

iii) Christmas tree

Q1. Given a no. represented in form of an array A, and an ip B, return $A \% P$.

Constraints

$1 < P \leq 10^9$

$1 \leq N \leq 10^5$.

$$A = \boxed{1 \mid 2 \mid 3 \mid 4 \mid 4}$$

⟹ 12344

$P = 4$

$44 \% 4 = 0$

$A \% P \Rightarrow 0$

$10^5$ elements ⟹ 9 9 9 9 9 9 9 9 9 . . . _ _ _ _ _ _ _ _ _ $10^6$ times

| 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 3 | 8 | 4 | 3 | 6 | 8 | 9 |

ex ⟹ 3 8 4 3 6 8 9 ⟹ 3843689

$3 \times 10^6 + 8 \times 10^5 + 4 \times 10^4 + 3 \times 10^3 + 6 \times 100 + 8 \times 10 + 9$

$A \Rightarrow 3\times10^6 + 8\times10^5 + 4\times10^4 + 3\times10^3 + 6\times10^2 + 8\times10 + 9$

$A\%P \Rightarrow \left[3\times10^6 + 8\times10^5 + 4\times10^4 + 3\times10^3 + 6\times10^2 + 8\times10 + 9\right]\%P$

additive rule $\Rightarrow (A+B)\%M \Rightarrow (A\%M + B\%M)\%M$

$$\Rightarrow \Big[(3\times10^6)\%P + (8\times10^5)\%P + (4\times10^4)\%P + (3\times10^3)\%P + (6\times10^2)\%P + (8\times10)\%P + 9\%P\Big]\%P$$

$\boxed{\text{multiplicative rule} \Rightarrow (a\times b)\%M \Rightarrow (a\%M \times b\%M)\%M}$

$$\Rightarrow \Big[(3\%P \times 10^6\%P)\%P + (8\%P\times10^5\%P)\%P + (4\%P\times10^4\%P)\%P + (3\%P\times10^3\%P)\%P + (6\%P\times10^2\%P)\%P + (8\%P\times10\%P)\%P + 9\%P\Big]\%P$$

| 3 | 8 | 4 | 3 | 6 | 8 | 9 |
|---|---|---|---|---|---|---|
| $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ |
| $10^6$ | $10^5$ | $10^4$ | $10^3$ | $10^2$ | $10$ | $1$ |

$(\underbrace{3\%P}_{\text{element of array digit in no.}} \times \underbrace{10^6\%P}_{\text{power of 10 each position}})\%P \Rightarrow \overset{i}{=} (arr[i]\%P \times 10^{N-i-1}\%P)\%P$

| i | power | length(N) |
|---|---|---|
| 0 | → 6 | 7 |
| 1 | → 5 | 7 |
| 2 | → 4 | 7 |
| 3 | 3 | 7 |
| 4 | 2 | 7 |
| 5 | 1 | 7 |
| 6 | 0 | 7 |

| 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 3 | 8 | 4 | 3 | 6 | 8 | 9 |

$3 \times 10^6 + 8 \times 10^5 + 4 \times 10^4 + 3 \times 10^3 + 6 \times 100 + 8 \times 10 + 9$

$7-0-1 = 6$

$7-1-1 = 5$

$7-2-1 = 4$

$\boxed{i = N-i-1}$

power at index i

ith index term => $(arr[i] \% P \times 10^{N-i-1} \% P) \% P$

0 - 9

$10^{N-i-1}$

$(0-9) \% P$

or, $10^{N-i-1} \% P$

=> $a^n \% P$ [power fun]

$a = 10$

$n = N-i-1$

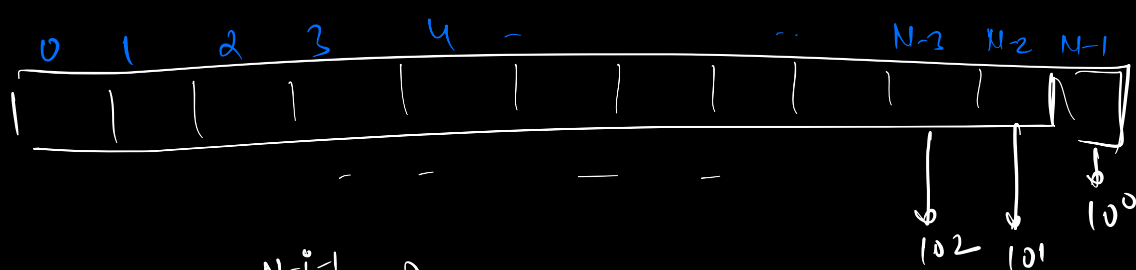
$P = P$

ith term => $(arr[i] \% P \times power(10, N-i-1, P)) \% P$

```
ans = 0
for (i=0; i<N; i++){
    ans = ans + (arr[i]%p x power(10, N-i-1, p))%p
}
return ans%p;
}
```

$O(N)$

$$\boxed{\begin{aligned} TC &\Rightarrow O(N^2) \\ SC &\Rightarrow O(1) \end{aligned}}$$




| $i$ | $10^{N-i-1} \% p$ |
|---|---|
| $N-1$ | $10^0 \% p \Rightarrow 1\% p$, if $p>1 = 1$, $p==1 = 0$ |
| $N-2$ | $10^1 \% p \Rightarrow (10 \times 10^0)\% p \Rightarrow (10\% p \times 10^0 \% p)\% p$ |
| $N-3$ | $10^2 \% p \Rightarrow (10 \times 10^1)\% p \Rightarrow (10\% p \times 10^1 \% p)\% p$ |
| $N-4$ | $10^3 \% p \Rightarrow (10 \times 10^2)\% p \Rightarrow (10\% p \times 10^2 \% p)\% p$ |
| $N-5$ | $10^4 \% p \Rightarrow (10 \times 10^3)\% p \Rightarrow (10\% p \times 10^3 \% p)\% p$ |
| ⋮ | |
| $0$ | $10^{N-1} \% p \Rightarrow (10 \times 10^{N-2})\% p = (10\% p \times 10^{N-2} \% p)\% p$ |

$c = 10 \% p$

$v = 1$

```
for( i= N-1; i >= 0; i-- ){
    ans = ans + ( arr[i]%p x v )%p
    v = (c x v )% p
}
return ans % p ;
```

$\boxed{\begin{array}{l} TC \Rightarrow O(N) \\ SC \Rightarrow O(1) \end{array}}$

**Q2. Alternating subarrays :**

Given a binary array, and a no. B, find and return mid indexes of alternating subarrays of len → $\underline{\underline{2B+1}}$.

$$\begin{array}{llccccc} & & 0 & 1 & 2 & 3 & 4 \\ ex \Rightarrow & A \Rightarrow & 1 & 0 & 1 & 0 & 1 \end{array}$$

$B \Rightarrow 1$

len $\Rightarrow 2B+1 = \underline{\underline{3}}$.

$\underline{O/P}$

ans ⇒ [1, 2, 3]

[1 0 1 0] ⇒ ✓

[0 1 0] ⇒ ✓

[1 0 1] ⇒ ✓

[1 0 0 1 0] ⇒ X X

[1] ⇒ ✓

[0] ⇒ ✓

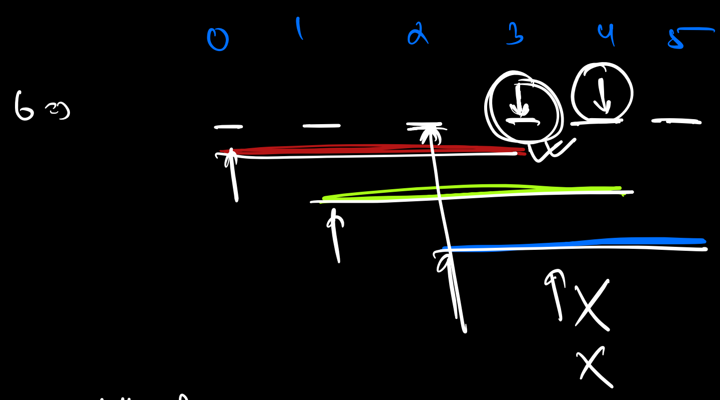
[0 0 1] ⇒ X X

N

subarrays ⇒ l

↓

4




$$\frac{N - l}{6 - 4}$$

```
l = 2B+1
ans = [ ]
for( i=0; i <= N-l ; i++ ){          => (N-l)
    prev = -1
    flag = 1

    for( j=i ; j < i+len ; j++ ){
        if( arr[j] == prev ){
            flag = 0;
            break;
        }
        prev = arr[j]
    }
    if( flag == 1 ){
        ans.push( i+B)
}
```

$$TC \Rightarrow (N-\ell)(\ell)$$
$$= (N-(2B+1))(2B+1)$$
$$= (N-2B-1)(2B+1)$$
$$\rightarrow \underline{\underline{N^2B}}$$

$$SC \Rightarrow O(N)$$

Christmas Tree

Q Given two arrays A & B

A represents size of christmas trees
B " cost of " " [respectively]

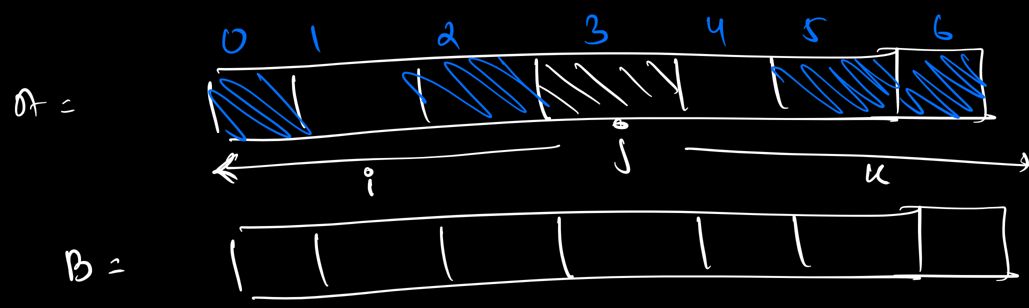
choose 3 trees i) $i < j < k$
ii) $A[i] < A[j] < A[k]$

minimise the overall cost iii) $B[i] + B[j] + B[k] \rightarrow$ minimal

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| A = | 1 | 2 | 4 | 6 |
| B = | 1 | 9 | 1 | 2 |

| idx ⇒ | 0 1 2 | 0 1 3 | 0 2 3 | 1 2 3 |
|---|---|---|---|---|
| A ⇒ | 1 2 4 | 1 2 6 | 1 4 6 | 2 4 6 |
| | 1+9+1 = 11 | 1+9+2 = 12 | 1+1+2 = 4 | 9+1+2 = 12 |

O/P = 4




① $i < j < k$

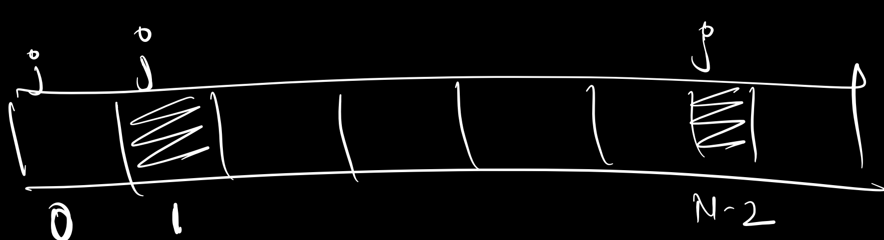
valid values for i are => all arr[i] < arr[j]

choose i out of valid values which least B[i]

cost => $\underset{\text{min}}{B[i]} + B[j] + B[k]$

valid values for k are => all arr[k] > arr[j]

choose k out of valid values which least B[k]



for ( j = 1 $\longrightarrow$ N-2) { $\longrightarrow$ N

min cost B[i] $\rightarrow$ N

min cost B[k] $\rightarrow$ N

pseudo

TODO

total B[i] + B[j] + B[k]

min

}

O( N * 2N)

TC $\Rightarrow$ O(N^2)

SC $\Rightarrow$ O(1)